BRUCE: Analytical exoplanet transit modelling and photometric analysis in Python and C

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Summary

A planet crossing its host star causes a small decrease in observed brightness. The shape and repetition of these transits constrain its relative size and orbit. Interpreting observations requires physical models, data preparation and parameter estimation. Repeated model evaluations become computationally demanding during fitting and searches of large datasets.

BRUCE is a Python and C package for analysing transiting exoplanets and related stellar systems. Its C modelling core is adapted from the Fortran-based `ellc` binary-star model (Maxted, 2016). It combines analytical transit and radial-velocity models with photometric processing, transit-template searches and parameter-estimation utilities. It also provides tools for retrieving observations from the Transiting Exoplanet Survey Satellite (TESS), investigating ambiguous periods and visualising orbital configurations. Its numerical routines are independently accessible. Source code and documentation are available in the BRUCE repository.

Statement of need

TESS (Ricker et al., 2015) has identified thousands of exoplanet candidates and produced extensive archives of light curves and full-frame images (Guerrero et al., 2021). Searching these data and fitting the resulting events require repeated model evaluations across many stars, epochs and trial parameters. The Next Generation Transit Survey (NGTS) (Wheatley et al., 2018) provides high-cadence survey photometry and extended monitoring of many long-period planet candidates from the TESS mission. Recovery campaigns (Gill et al., 2020b; Gill et al., 2020a; Gill et al., 2020c) and subsequent work (Gill et al., 2024) exemplify the large, often discontinuous datasets that must be searched for sparse transit signals. Gill et al. (2024) searched more than 200,000 archival NGTS measurements for a single target. Efficient models are therefore needed both to scan these observations and to fit events jointly with TESS data. PLATO's planned long-baseline observations of large stellar samples (Rauer et al., 2025) will extend the need for computationally efficient detection and characterisation tools.

BRUCE combines fast modelling with data retrieval, normalisation, binning, template searches and period-alias assessment. A fitted transit can become a search template, and the resulting events can constrain follow-up observations. This package processes many candidates efficiently while permitting users to adjust each analysis choice.

State of the field

Several established packages address parts of this problem. `batman` provides transit calculations for a range of limb-darkening laws (Kreidberg, 2015). `ellc` models detached eclipsing binaries and transiting planets with a broader treatment of stellar geometry and binary effects (Maxted, 2016). PyTransit provides efficient transit models and interfaces designed for demanding photometric analyses (Parviainen, 2015), including a documented power-2 implementation. Lightkurve provides an extensive environment for accessing and analysing Kepler and TESS observations (Lightkurve Collaboration, 2018).

BRUCE builds on these methodological foundations rather than introducing a new transit theory. Its contribution is the combination of a compact C implementation, optional shared-memory parallelism, direct model-and-likelihood evaluation and workflows for matching known transit shapes and investigating period ambiguity. The package focuses on speed and provides a single environment in which exoplanet researchers can find transit signals, fit planetary systems and analyse the supporting photometry.

Software design

The binary-star modelling core is adapted from `ellc` (Maxted, 2016), with an analytical eccentric-anomaly solver based on Markley (1995): a cubic approximation followed by a fifth-order correction solves the elliptic Kepler equation. The resulting true anomaly determines Keplerian radial velocities and the projected separation used for photometric transits. Transit evaluation focuses on the fast analytical `qpower2` approximation (Maxted & Gill, 2019), using the power-2 limb-darkening law. Avoiding repeated numerical integration over the stellar disc makes this approximation efficient for fitting exoplanet transits. Additional intensity profiles and integration routines exist in the C source, although the power-2 law is used by default.

Python supplies array-based interfaces, data handling and fitting orchestration, while C with OpenMP handles repeated calculations. NumPy arrays connect these layers; contiguous double-precision arrays retain numerical precision and avoid unnecessary conversions. OpenMP distributes sample-wise calculations across CPU threads, further accelerating fitting and minimisation routines. A combined evaluation mode accumulates a likelihood statistic without returning the full model array, although its performance advantage depends on workload and memory costs. Numerical cadence integration requires odd subsample counts of at least three, with a default of 11, centred across each exposure. Keplerian radial velocities support primary and double-lined systems, and sampling helpers interface with `emcee` (Foreman-Mackey et al., 2013) and SciPy-based minimisation routines (Virtanen et al., 2020).

The higher-level modules support TESS retrieval, photometric time-series preparation, event fitting, template matching and period-alias assessment. Detrending remains an explicit modelling choice: this package provides filter routines that can remove stellar variability but also distort a transit if not used appropriately. Separating the normalisation from the transit template

matching allows an inspection of that dependency. Plotting, transit-timing helpers and orbit animations support interpretation of fitted systems. Figure 1 compares the analytical transit model with independent numerical integration.

Use cases

Direct transit fitting. BRUCE provides fitting routines and sampling utilities that simplify preparing data, fitting transit models and inspecting the results. Supplying photometry and uncertainties to `bruce.binarystar.lc` evaluates the model and accumulates the likelihood statistic in C, avoiding transfer of a model array at every optimiser call or sampler step. The current wrapper returns negative log likelihood, so posterior sampling requires the corresponding sign conversion.

Detecting single transits. Transit templates can be compared to photometric data over a bank of trial epochs to support blind searches of transit events. Published recovery examples include NGTS-11 b (Gill et al., 2020c) and TOI-2447 b/NGTS-29 b (Gill et al., 2024). The latter uses a TESS-derived template to recover events in archival NGTS survey observations and subsequent monitoring. These methods apply to both ground-based and space-based photometry, including searches through TESS data for additional transits of long-period planetary systems. Such searches can be computationally expensive, but BRUCE provides template matching and normalisation tools to support them. Detection thresholds must be calibrated for each dataset and search; this calibration will be explored in future publications.

Resolving duotransits. Two separated transits permit discrete aliases $P_n = (t_2 - t_1)/n$. BRUCE provides a period-resolution framework that organises transit events, generates candidate aliases and combines event fitting, observing coverage and searches for additional events to assess the possible periods. This framework simplifies the analysis of duotransit systems such as those identified in TESS samples (Hawthorn et al., 2024).

Model comparison

Following the geometry of Figure 3 in Maxted & Gill (2019), BRUCE is compared with `ellc` numerical integration using identical power-2 coefficients (Maxted, 2018). No parameters are fitted. For this case, the maximum difference is 74.0 ppm; changing `ellc` from fine to very-fine grids changes the reference by at most 0.58 ppm. Source code and model arrays accompany the paper.

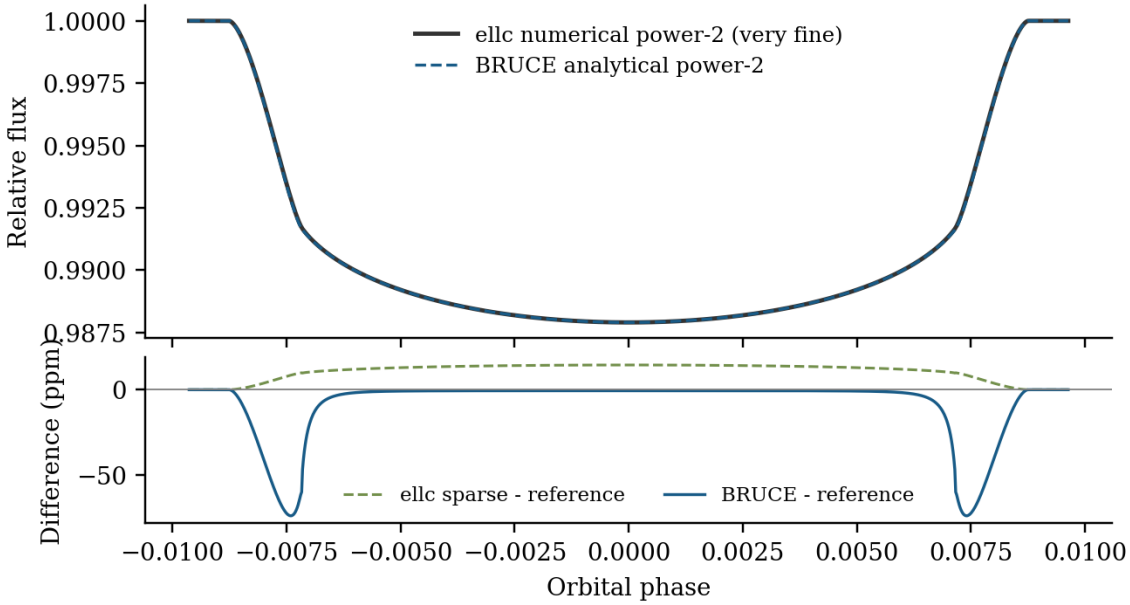


Figure 1. Analytical versus numerical power-2 transit models for spherical bodies with scaled stellar radius 0.05, radius ratio 0.1, zero impact parameter and eccentricity, and instantaneous sampling. Both use c=0.715452 and alpha=0.643777. Upper: BRUCE and very-fine ellc integration. Lower: differences from that reference, including sparse ellc integration for context. This matched-law comparison follows the layout of Maxted & Gill (2019), Figure 3; it does not reproduce their STAGGER-profile reference.

Computational performance

BRUCE is a fast light-curve framework and its speed is demonstrated with a benchmark test that records model-only calls and direct C likelihood calls with `flux` and `flux_err` supplied to `bruce.binarystar.lc`. Figure 2 shows the latter, the relevant repeated evaluation for fitting. Contiguous double-precision arrays contain 100 to 100,000 samples; seven batches follow warm-up in separate one- and four-thread processes. The instantaneous circular case uses e=0, the instantaneous eccentric case e = 0.4, and the exposure-integrated case e=0 with 11 subsamples over 2 minutes. All use fixed uncertainties, zero jitter and `offset=0`, omitting variance-dependent normalisation. Imports, input construction, I/O and optimiser execution are excluded; settings, raw timings, source hashes and the compiler log are retained.

On an Intel Core i7-9750H at a nominal 2.60 GHz, with six physical cores, Python 3.9.13, NumPy 1.26.4 and Apple Clang 21.0.0 with OpenMP, the 1,000-sample circular likelihood achieved approximately 7,500 evaluations per second with one thread and 28,000 with four. For 100,000 samples, the rates were 57 and 280 evaluations per second, a speed-up of 4.90. At that size, the one-thread eccentric and exposure-integrated rates were 25.6 and 11.0 evaluations per second. Rates describe complete model-and-likelihood evaluations, not independent posterior samples.

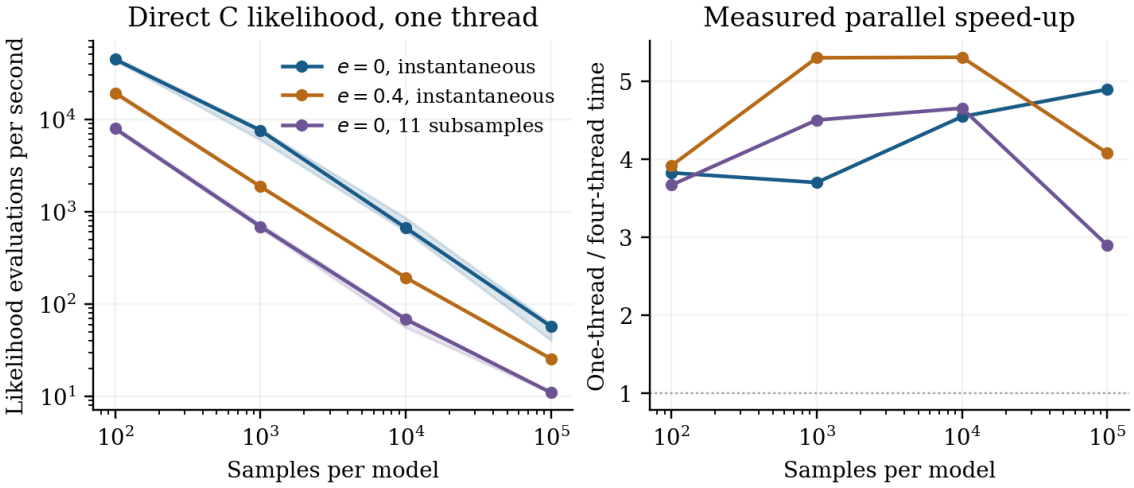


Figure 2. Direct C likelihood throughput and OpenMP scaling for complete model evaluations. Left: reciprocal median call time with one thread; shading spans reciprocals of the upper and lower timing quartiles. Right: one-thread divided by four-thread median call time. Labels specify eccentricity and exposure treatment. The same retained benchmark data underpin both panels; these are local BRUCE measurements, not comparisons with competing packages.

The C extension used by BRUCE (`bruce_c`) is checked against a model returned to Python and reduced with NumPy. It avoids allocating an output model but is not uniformly faster in the measured cases. These tests are hardware dependent; multiprocessing analyses should control OpenMP threads to avoid oversubscription. Users should be aware of the `OMP_NUM_THREADS` environment variable to control the number of threads used when calculating a log-likelihood or generating a model. While OpenMP is resilient, users should set it to 1 when fitting data with multiprocessing (e.g. with emcee).

Research impact statement

BRUCE grew from the need to search and fit large photometric datasets efficiently, particularly TESS and NGTS long-period candidates. The author's recovery programme includes TIC-238855958, TIC-231005575 and NGTS-11 b (Gill et al., 2020b; Gill et al., 2020a; Gill et al., 2020c), followed by TOI-2447 b/NGTS-29 b (Gill et al., 2024); the TESS duotransit catalogue of Hawthorn et al. (2024) further motivates tools for ambiguous periods and targeted monitoring. These studies motivate BRUCE's workflows; they do not all use the present release.

BRUCE provides undergraduate and postgraduate students, as well as established researchers, with tools to find, fit and prepare planetary-system analyses for publication. The educational and research scope of BRUCE lowers the amount of processing needed to move between candidate detection, parameter estimation and follow-up planning. Reproducible tests, benchmarks and regression checks are provided.

AI usage disclosure

OpenAI Codex assisted with manuscript drafting, bibliography preparation, benchmark and figure scripts, documentation, and preceding numerical refactoring and regression-test development. The author subsequently edited the text substantially and checked that the text and figures reflect the intended content. The author used Codex assistance for the NumPy C API bindings. Automated checks included compilation and regression tests.

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